Lab Sheet 5
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```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from skimage.feature import hog
from skimage import exposure
from google.colab import files
```

```
image_path = "/content/Landscape.png"

print("Image Path:", image_path)
```

```
Image Path: /content/Landscape.png
```

```
image = cv2.imread(image_path)

if image is None:
    print("Image not found. Check the image path.")
else:
    print("Image loaded successfully.")
```

```
Image loaded successfully.
```

```
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(8, 6))
plt.imshow(image_rgb)
plt.title("Original Image")
plt.axis("off")
plt.show()
```




```
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

plt.figure(figsize=(8, 6))
plt.imshow(gray, cmap="gray")
plt.title("Grayscale Image")
plt.axis("off")
plt.show()
```




```
gray_blur = cv2.GaussianBlur(gray, (3, 3), 0)

plt.figure(figsize=(8, 6))
```

```
plt.imshow(gray_blur, cmap="gray")
plt.title("Preprocessed Image")
plt.axis("off")
plt.show()
```




```
sift = cv2.SIFT_create()

keypoints, descriptors = sift.detectAndCompute(gray_blur, None)

print("Number of SIFT Keypoints:", len(keypoints))
print("Descriptor Shape:", descriptors.shape)
```

```
Number of SIFT Keypoints: 6111
Descriptor Shape: (6111, 128)
```

```
sift_image = cv2.drawKeypoints(
    image_rgb,
    keypoints,
    None,
    flags=cv2.DRAW_MATCHES_FLAGS_DRAW_RICH_KEYPOINTS
)

plt.figure(figsize=(10, 7))
plt.imshow(sift_image)
plt.title("SIFT Keypoints")
plt.axis("off")
plt.show()
```




```
gray_float = gray_blur / 255.0

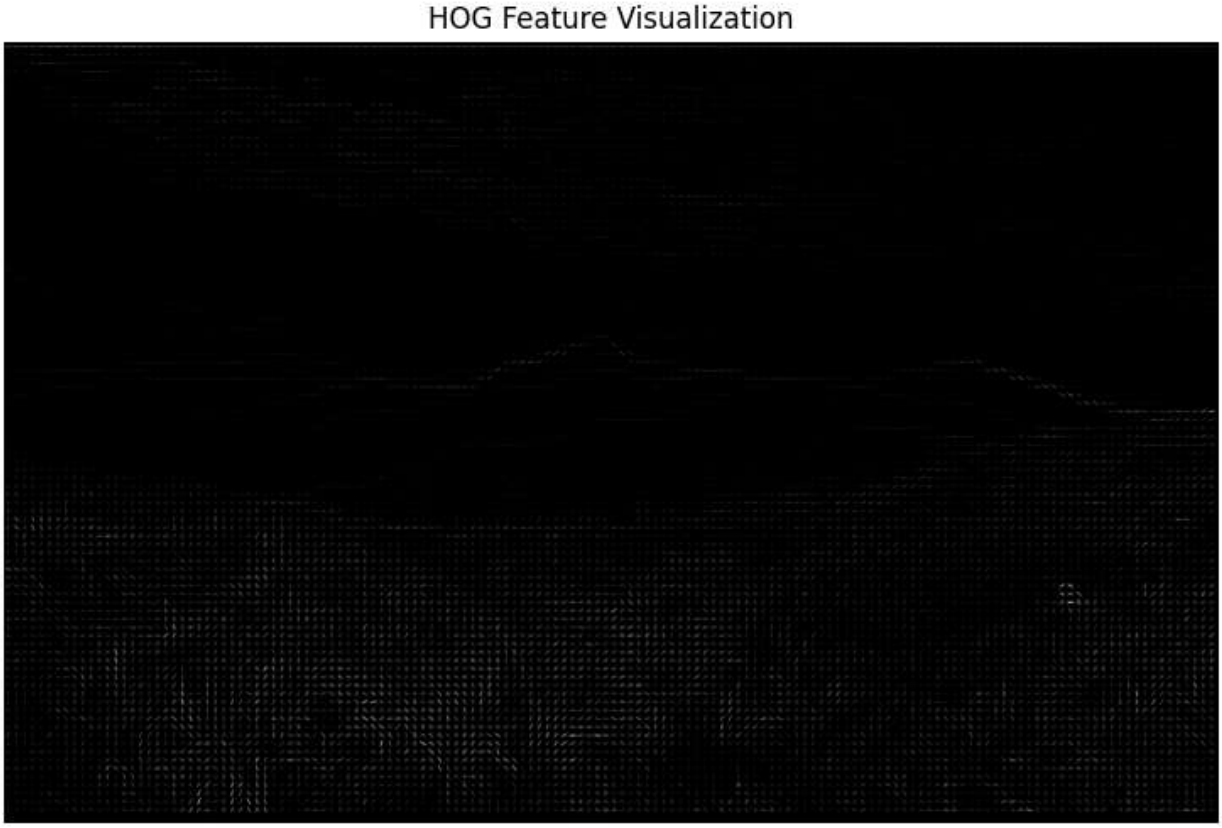
hog_features, hog_image = hog(
    gray_float,
    orientations=9,
    pixels_per_cell=(8, 8),
    cells_per_block=(2, 2),
    visualize=True
)

print("Number of HOG Features:", len(hog_features))
```

```
Number of HOG Features: 485460
```

```
hog_image_rescaled = exposure.rescale_intensity(
    hog_image,
    in_range=(0, 10)
)

plt.figure(figsize=(10, 6))
plt.imshow(hog_image_rescaled, cmap="gray")
plt.title("HOG Feature Visualization")
plt.axis("off")
plt.show()
```




```
image2_path = "/content/landscape 2.png"

print("Second Image Path:", image2_path)
```

```
Second Image Path: /content/landscape 2.png
```

```
img1 = cv2.imread(image_path, cv2.IMREAD_GRAYSCALE)
img2 = cv2.imread(image2_path, cv2.IMREAD_GRAYSCALE)

if img1 is None or img2 is None:
    print("One or both images could not be loaded.")
else:
    print("Both images loaded successfully.")
```

```
Both images loaded successfully.
```

```
sift = cv2.SIFT_create()

kp1, des1 = sift.detectAndCompute(img1, None)
kp2, des2 = sift.detectAndCompute(img2, None)

bf = cv2.BFMatcher()

matches = bf.knnMatch(des1, des2, k=2)

good_matches = []

for m, n in matches:
    if m.distance < 0.75 * n.distance:
        good_matches.append(m)
```

```
print("Number of Good Matches:", len(good_matches))
```

```
Number of Good Matches: 20
```

```
matched_image = cv2.drawMatches(
    img1, kp1,
    img2, kp2,
    good_matches,
    None,
    flags=cv2.DrawMatchesFlags_NOT_DRAW_SINGLE_POINTS
)

plt.figure(figsize=(14, 7))
plt.imshow(matched_image, cmap="gray")
plt.title("SIFT Feature Matching")
plt.axis("off")
plt.show()
```

